Minju Hong

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RESEARCH INTERESTS

Theoretical foundations of RLHF; preference learning and pluralistic LLM alignment; optimization dynamics in games; statistical learning theory

AFFILIATION

Carnegie Mellon University — *May. 2026 – Present*
Research Intern, Language Technologies Institute — Pittsburgh, PA, USA

Korea Advanced Institute of Science and Technology (KAIST) — *Mar. 2024 – Feb. 2026*
M.Sc. in Electrical Engineering — Daejeon, Korea
Advisor: Prof. Hye Won Chung

Korea Advanced Institute of Science and Technology (KAIST) — *Mar. 2019 – Feb. 2024*
B.Sc. in Mathematical Sciences & Electrical Engineering (*Double Major*) — Daejeon, Korea
Graduated with Excellence in Leadership and Volunteer Activity

Technical University of Munich — *Apr. 2023 – Sep. 2023*
Exchange Student in Mathematics — Munich, Germany

PUBLICATIONS

[1] Provably Efficient Regularized Online RLHF with Generalized Bilinear Preferences
Junghyun Lee, **Minju Hong**, Kwang-Sung Jun, Chulhee Yun, Se-Young Yun.
ICML 2026 Pluralistic Alignment Workshop
CKAIA 2026 Outstanding Paper Award

SELECTED RESEARCH PROJECTS

Modeling Subjective Annotations in LLM Alignment — *May 2026 – Present*
Research Intern, Carnegie Mellon University, Language Technologies Institute — Pittsburgh, PA
Supervised by Prof. Maarten Sap; mentored by Dr. Vasudha Varadarajan

- Investigating methods for representing subjective human annotations using probabilistic methods.

Implicit Bias of Softmax Dynamics in Bilinear Zero-Sum Games — *Mar. 2026 – Present*
Research collaboration with the OptiML Lab, KAIST AI; with Prof. Chulhee Yun — Seoul, Korea

- Characterizing the learning dynamics and implicit regularization of policies.

Regularized Online RLHF with Generalized Bilinear Preferences — *Sep. 2025 – May. 2026*
M.S. thesis research and continued collaboration — Daejeon, Korea

- Studied online RLHF under generalized bilinear preference models that capture pairwise interactions beyond scalar reward models.
- Contributed to regret analyses for regularized self-play algorithms and statistical complexity analyses for bilinear preference learning.
- Coauthored a paper presented at the ICML 2026 Workshop on Pluralistic Alignment.

RESEARCH EXPERIENCE

Graduate Research Assistant — *Mar. 2024 - Feb. 2026*
Advised by Prof. Hye Won Chung — Daejeon, Korea

- Investigated the theoretical foundations of order sensitivity in In-Context Learning (ICL) in Large Language Models.
- Analyzed the causal mechanisms behind order sensitivity from an optimization perspective.

Individual Study: Information Spread Dynamics — *Apr. 2023 - Sep. 2023*
Undergraduate Researcher, Advised by Prof. Debarghya Ghoshdastidar — Munich, Germany

- Studied misinformation mitigation strategies within the Theoretical Foundations of AI Group.
- Developed and evaluated blocker selection algorithms to minimize the spread of misinformation on graph networks.

Project: Mathematical Biology (MAS467) *Sep. 2023 - Dec. 2023*
Course Term Project, Advised by Prof. Jaekyung Kim Daejeon, Korea
Title: Shedding Light on Combined Multiple Transcriptional Repression Mechanisms

- Modeled cell cycle dynamics to analyze the stability of transcriptional repression mechanisms.
- Conducted simulations to observe the interaction between multiple repression pathways.

Individual Study: Random Graph Matching *Mar. 2021 - Dec. 2022*
Undergraduate Researcher, Advised by Prof. Hye Won Chung Daejeon, Korea

- Conducted independent study on random graph theory and delivered literature review presentations.
- Focusing on random graph matching algorithms, investigated better algorithmic strategies for stochastic block models.

HONORS AND AWARDS

Doo-Myeong Undergraduate Scholarship *Spring 2020*
Awarded to one student from the Department of Mathematics, Class of 2023
A merit-based scholarship granted by the Miwon Group.

National Science and Engineering Undergraduate Scholarship *Spring 2019*
Merit-based scholarship awarded by Korea Student Aid Foundation.

Leadership Mileage Certificate, KAIST *Fall 2020*
Honor for top 3% among 3,600+ students with top achievements in leadership activities including volunteering and campus activities.

Grand Prize, World Friends Korea ICT Volunteers Award *Fall 2019*
Won by the KAIST-Tanzania team for outstanding ICT volunteer projects.

TEACHING EXPERIENCE

Machine Learning Basics and Practices (EE214) *Fall 2025*
Teaching Assistant, KAIST

Probability and Introductory Random Processes (EE210) *Spring 2025*
Teaching Assistant, KAIST

EE Career Design (EE215) *Spring and Fall 2025*
Teaching Assistant, KAIST; provided simultaneous Korean–English interpretation for an international course.

SERVICE AND LEADERSHIP

Graduate Student Representative, KAIST School of Electrical Engineering *Aug. 2024 – Feb. 2025*
Organized academic talks and community events for more than 400 graduate students.

Student Council Representative, KAIST Department of Mathematical Sciences *Mar. 2021 – Feb. 2023*
Organized academic events and seminars for more than 100 undergraduate students.

TECHNICAL SKILLS

Programming Languages: Python, MATLAB, LaTeX

LANGUAGES

English Highly proficient (iBT TOEFL 117/120 : R30/L30/S30/W27)
Korean Native